

Lab 1: Version 1

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Version 1

Table of Contents

1 Introduction.....	3
2 Product Description.....	4
2.1 Key Product Features and Capabilities.....	5
2.2 Major Components (Hardware/Software).....	6
3 Identification of Case Study.....	7
3.1 Case Study Participants.....	7
3.2 Current Referral Process and Challenges.....	8
3.3 Features Under Test.....	9
3.4 PatientPath Referral Workflow.....	10
3.5 Future Applications.....	11
Glossary.....	12
References.....	13

List of Figures

Figure 1: PatientPath Record Exchange Flow

List of Tables

No tables included in this draft.

1 Introduction

Healthcare organizations still have problems sharing patient information between different systems. Fax is still used for many medical record transfers, which can make the process slow and difficult to track. The Office of the National Coordinator for Health Information Technology reported that about 65% of medical record transfers still used fax in 2021 (Office of the National Coordinator for Health Information Technology [ONC], 2023). Other reports also show that many providers have difficulty exchanging information with outside organizations (ONC, 2022).

These problems can cause delays for providers and patients. Staff may have to resend records, check whether a referral was received, or manually keep track of paperwork. Missing information can also lead to repeated tests and other unnecessary work. These problems are especially important for smaller and rural healthcare organizations that may have fewer technical resources.

PatientPath is a proposed solution to these problems. It is a web-based system that helps healthcare organizations share records and manage referrals. Instead of replacing an existing electronic health record system, PatientPath is designed to work between existing systems. The main goals are to make record sharing faster, make referrals easier to follow, and keep patient information protected. The system also includes features for organizations with limited or unreliable internet connections.

2 Product Description

PatientPath is a secure web-based platform for sharing medical records and managing referrals between healthcare organizations. It is intended to reduce the manual work involved in faxing records while allowing organizations to keep their existing electronic health record (EHR)

systems. A provider can create a referral, identify the records that need to be shared, and monitor the referral through its status. The receiving organization can review the shared records and update the referral status, while administrative users can monitor the process without needing access to clinical information. Patients can also manage consent for sharing their information.

2.1 Key Product Features and Capabilities

PatientPath incorporates several features designed to improve the security, efficiency, and reliability of electronic record exchange between healthcare organizations. The system provides secure digital record exchange, interoperability with existing electronic health record (EHR) systems, automated referral management, access and security controls, and support for environments with unreliable network connectivity.

The secure record exchange functionality allows authorized users to upload, access, and transfer medical records electronically rather than relying on traditional fax-based workflows. By providing a centralized mechanism for transferring records, PatientPath is designed to reduce problems associated with manual transmission, including lost pages, illegible documents, and delays in receiving patient information. This capability supports more efficient communication between referring providers and receiving specialists while maintaining appropriate access controls.

PatientPath also supports interoperability through the use of FHIR and HL7 healthcare data standards. These standards allow the system to exchange and organize information across different EHR environments without requiring healthcare organizations to replace their existing systems. By operating between existing EHR platforms, PatientPath is intended to provide a consistent method for exchanging patient information across organizational and system boundaries.

The platform further incorporates an automated referral workflow and status-tracking system. Providers can initiate referrals, identify the records required for the referral, authorize access for the receiving organization, and monitor the referral as it progresses through different stages. Automated notifications can identify referrals that have not been updated within a defined period, reducing the need for administrative staff to rely on repeated phone calls, manual record requests, and other forms of follow-up. This functionality provides greater visibility into the referral process and helps organizations identify delays more efficiently.

Security and access control are integrated throughout the PatientPath platform. Patient consent management allows patients to control the sharing of their health information, while role-based access control limits users to the information and functions required for their responsibilities. Multi-factor authentication provides an additional layer of account security, and encryption is used to protect sensitive information. PatientPath also maintains audit records of relevant access and transfer activity, providing organizations with a record of how information is accessed and exchanged.

Because rural and lower-resource healthcare organizations may experience unreliable network connectivity, PatientPath also incorporates offline synchronization capabilities. When connectivity is temporarily unavailable, the system can cache required information locally and synchronize changes once the connection is restored. This functionality is intended to allow essential referral and record-sharing activities to continue despite temporary network interruptions.

Together, these capabilities provide PatientPath with a unified workflow for secure, cross-organizational record exchange and referral coordination. Rather than replacing an organization's existing EHR, PatientPath operates alongside existing systems to facilitate the

secure transfer of information, provide visibility into referral status, and support healthcare organizations with different levels of technical and network infrastructure.

2.2 Major Components (Hardware/Software)

PatientPath does not require special hardware for users. The system can be accessed from a normal computer, tablet, or mobile device using a web browser. The production system would run on cloud infrastructure and would use servers, databases, encrypted storage, and a key management service. Patient information would be stored in protected systems rather than on a user's personal computer.

The software uses a three-layer structure. The Presentation Layer contains the user interfaces, including the provider dashboard, patient portal, administrator area, and login screens. The Application Layer contains the main services that handle record exchange, user access, patient records, and FHIR and HL7 integration. The Data Layer stores medical documents, user information, normalized healthcare data, audit records, and consent information.

The planned software uses React and TypeScript for the front end and Node.js with Express for the back end. PostgreSQL is used for the main database and Prisma is used to communicate with it. The system also uses WebSockets for status updates and TOTP for multi-factor authentication. Encryption and cloud key management are used to protect information. GitHub and GitHub Actions can be used for source control and development automation.

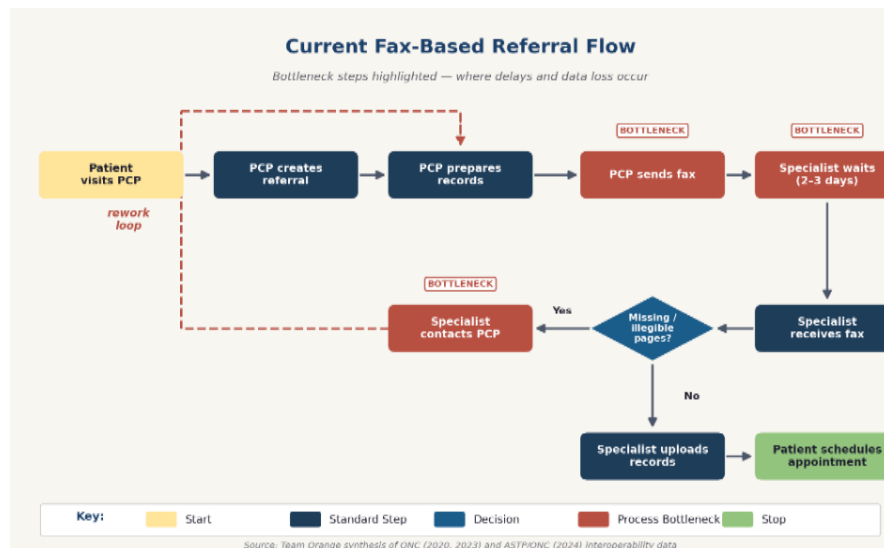
Figure 1*PatientPath Record Exchange Flow*

Figure 1 shows the basic idea of how PatientPath can move a referral and patient record between organizations while keeping the user's access and consent controls in place.

3 Identification of Case Study

3.1 Case Study Participants

Our case study focuses on a referral between a rural primary care clinic and a regional hospital. Dr. Sarah Chen is a primary care physician at a mid-sized rural clinic with limited information technology resources. Her patient, Robert Mills, is a 67-year-old who has recently been diagnosed with a cardiac condition that requires evaluation by a specialist. The receiving specialist is Dr. James Park, a cardiologist at a regional hospital located approximately 45 miles from Dr. Chen's clinic. Linda Torres, an administrative coordinator at the clinic, assists with managing the referral and monitoring its progress.

This case study represents a common situation in which healthcare organizations use different electronic health record (EHR) systems and have limited interoperability between them. The rural clinic uses an older EHR system that cannot directly exchange records with the hospital's system. As a result, the clinic currently relies on manual processes to transfer patient information to the specialist.

3.2 Current Referral Process and Challenges

Under the current process, Dr. Chen must prepare Robert's referral and provide the receiving hospital with the relevant medical records. Because the clinic's EHR cannot directly exchange information with the hospital's system, staff members must send the records by fax. The referral may contain more than 15 pages of medical information, increasing the possibility of missing or illegible pages during transmission. After receiving the referral, Dr. Park's office may need to contact the clinic to request missing information or confirm that the complete record was received.

This process can delay Robert's access to specialized care. In the case study, the specialist may wait two to three days to obtain the necessary records, resulting in Robert's appointment being delayed by approximately one week. The clinic also experiences occasional internet connectivity interruptions, which can make cloud-based systems difficult to use during periods of downtime. These challenges demonstrate the need for a system that can facilitate secure information exchange between different healthcare organizations while continuing to support providers in environments with unreliable connectivity.

3.3 Features Under Test

Our case study will demonstrate how PatientPath addresses the primary challenges in this referral process. PatientPath's secure digital record exchange and FHIR/HL7 interoperability capabilities will allow Dr. Chen's clinic to electronically transfer the required patient records between its existing EHR environment and the receiving hospital's system without relying on fax transmission. The automated referral workflow will allow Dr. Chen to create the referral, select the appropriate patient records, and authorize access for the receiving organization. These capabilities are intended to provide a more efficient and consistent method for exchanging information across different healthcare environments.

PatientPath will also use consent management and role-based access control to ensure that patient information is available only to appropriately authorized users. Robert's consent will determine whether his records can be shared, while users such as Dr. Park and Linda will receive access based on their roles within the referral process. Linda, for example, will be able to manage referral information without receiving unrestricted access to clinical records. PatientPath will maintain an audit trail of relevant record access and transfer activity to support accountability and review. Finally, the offline synchronization capability will allow the clinic to continue working during a temporary connectivity interruption and synchronize information once the connection is restored.

3.4 PatientPath Referral Workflow

The proposed workflow begins when Dr. Chen creates a referral for Robert through PatientPath. The system will prompt her to identify the records required by the receiving specialist and confirm that the appropriate patient consent has been obtained. Dr. Chen can then

select the relevant records and authorize the receiving organization to access the information required for the referral. PatientPath will securely transmit the records and notify Dr. Park's organization that the referral is available.

After receiving the referral, Dr. Park will be able to access the authorized records through PatientPath and review the information needed for Robert's care. Linda will be able to monitor the referral status and identify whether additional action is required without having unrestricted access to Robert's clinical information. PatientPath will record significant actions throughout the process, including record access and transfers, providing an audit trail for later review. If the clinic experiences a temporary loss of internet connectivity, PatientPath will allow supported work to continue using locally cached information and synchronize changes when connectivity is restored.

The intended outcome is to reduce the time required to exchange records and eliminate unnecessary manual fax-based steps. In this scenario, PatientPath is designed to allow Dr. Park to receive a complete and legible set of authorized records without waiting several days for fax transmission or requesting missing pages. This can allow Robert's referral to be processed more efficiently and potentially reduce the delay between his initial diagnosis and specialist appointment.

3.5 Future Applications

Although this case study focuses on a rural primary care clinic and a regional cardiology practice, PatientPath is designed to support additional healthcare organizations and referral scenarios. Future applications could include larger hospital networks, multi-site healthcare organizations, behavioral health providers, long-term care facilities, specialty practices, and

federally qualified health centers. Expanding PatientPath to these environments would allow the same secure record-sharing and referral-tracking capabilities to be applied across a wider range of healthcare workflows.

Future development could also include a patient portal that allows patients to participate more directly in record-sharing and consent processes, as well as integrations with telehealth services. PatientPath could additionally explore integration with health information exchange networks such as CommonWell to support broader information sharing between participating healthcare organizations. These future applications would build on the core PatientPath architecture while expanding its ability to support interoperability and referral coordination across different healthcare settings.

Glossary

EHR: Electronic Health Record. A digital record of a patient's healthcare information.

FHIR: Fast Healthcare Interoperability Resources. A standard used to exchange healthcare information between systems.

HL7: Health Level Seven. A group of healthcare data standards used to exchange information.

RBAC: Role-Based Access Control. A method of giving users access based on their role.

MFA: Multi-Factor Authentication. A login method that uses more than one form of verification.

API: Application Programming Interface. A way for different software systems to communicate.

HIE: Health Information Exchange. A system or organization that supports the exchange of healthcare information.

FQHC: Federally Qualified Health Center. A type of community healthcare provider that receives federal support.

HIPAA: Health Insurance Portability and Accountability Act. U.S. requirements related to protecting health information.

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